Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Course : B.Tech
Code : 102601

Subject: Design of Machine Elements

Time: 03 Hours
Full Marks: 70

Instructions:-
(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct answer the following (Any seven questions only):*** [2 x 7 = 14]

(a) The lowest value of joint efficiency is assumed in the case of
(i) single riveted butt joint
(ii) double riveted butt joint
(iii) double riveted lap joint
(iv) single riveted lap joint

(b) Standardization deals with the characteristics of products that include
(i) dimensions of machine elements
(ii) composition and properties of engineering materials
(iii) method of testing the product
(iv) all the three

(c) The gearbox is produced by
(i) design by drawing
(ii) design by craft evolution
(iii) design synthesis
(iv) simultaneous design

(d) Steels used for welded assemblies are
(i) medium carbon steel
(ii) high carbon steel
(iii) mild steel
(iv) alloy steel

(e) In a shaft-basis system, the basis shaft is one
(i) whose upper deviation is zero
(ii) whose upper and lower deviations are zero
(iii) whose lower deviation is zero
(iv) none of the above

(f) According to Indian standards, the total number of tolerance grades are
(i) 10
(ii) 18
(iii) 20
(iv) 8

(g) The shock-absorbing capacity of a bolt can be increased by
(i) increasing shank diameter
(ii) using castle nut in place of an ordinary hexagonal nut
(iii) making the shank diameter equal to the core diameter of the threads
(iv) using a washer

(h) The maximum stress concentration factor for a rectangular plate with a transverse hole loaded in tension or compression is
(i) 2
(ii) 3
(iii) 2.5
(iv) 1

(i) The reliability factor for using 50% reliability in design is
(i) 0.897
(ii) 0.868
(iii) 1
(iv) 0.814

(j) The endurance limit of the component can be increased by
(i) increasing the size of the component
(ii) shot peening
(iii) increasing the stress concentration
(iv) coating

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Q.2 (a) What are adaptive design and optimum design? Describe some factors that influence machine design. [7]

(b) Define the factor of safety. What is its importance in design? What are various factors to be considered in deciding the factor of safety? [7]

Q.3 (a) Explain notch sensitivity. State the relation between stress concentration factor, fatigue stress concentration factor and notch sensitivity. [7]

(b) Describe various types of loads that machine elements are subjected to and specify the factor of safety uses in each case. [7]

Q.4 (a) Explain briefly a design of shafts subjected to combined bending and torsion. [7]

(b) Name the various standards used in the design of machine elements. Calculate the tolerance, fundamental deviation and limits of sizes for the shaft designated as 70 H8–g6. [7]

Q.5 (a) Explain briefly the design considerations of welded assemblies. How the strength of the transverse fillet weld is evaluated? [7]

(b) A welded rectangular steel tube is 1000 mm long, has a 3 mm wall thickness, and a 60 mm by 90 mm rectangular cross-section, as shown in Fig. 1. Assume an allowable shear stress of 80 MPa and a shear modulus of 80 GPa. [7]

(a) Estimate the allowable torque T.

(b) Estimate the angle of twist due to the torque.

Q.6 Mention the utility of the centre bolt, U- clamp, rebound clip and camber in a leaf spring. Design a leaf spring for the following specifications: Total load = 140 kN; Number of springs supporting the load = 4; Maximum number of leaves = 10; Span of the spring = 1000 mm; Permissible deflection = 80 mm. Take Young's modulus, E = 200 kN/mm^2 and allowable stress in spring material as 600 MPa. [14]

Q.7 (a) Explain with sketches the different types of failures and efficiencies of the riveted joints. [7]

(b) Discuss the design procedure for the eccentrically loaded bolted joint. [7]

Q.8 Figure 2 depicts a band brake whose drum rotates counterclockwise at 200 rev/min. The drum diameter is 400 mm, and the band lining is 75 mm wide. The coefficient of friction is 0.20. The maximum lining interface pressure is 480 kPa. [14]

(a) Find the brake torque, necessary force P, and steady-state power.

(b) Complete the free-body diagram of the drum. Find the bearing radial load that a pair of straddle-mounted bearings would have to carry.

(c) What is the lining pressure p at both ends of the contact arc?

Q.9 A line shaft is to transmit 30 KW at 160 rpm. It is driven by a motor placed directly under it by means of a belt running on a 1m diameter pulley keyed to the end of the shaft. The tension in the tight side of the belt is 2.5 times that of the slack side, and the centre of the pulley overhangs 150 mm beyond the centre line of the end bearing. Determine the diameter of the shaft if the allowable shear stress is 56MPa and the pulley weighs 1690 N. [14]

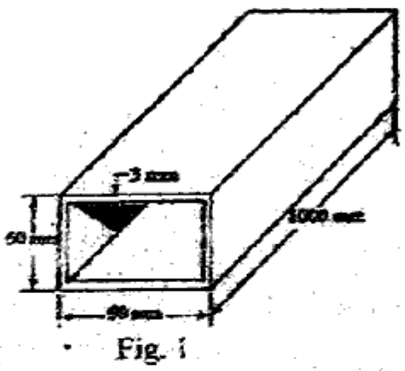


Fig. 1

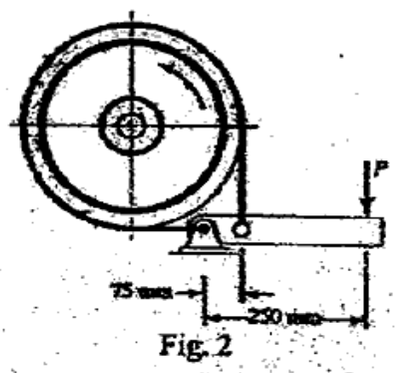


Fig. 2

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